

Lesson 19 — Multiple Variables & Data You Can Reason About

Matches LEGO Teacher Portal lesson: "Feelings in Flow".

Objective (I can...): I can use several variables together and reason about relationships between them.

CS concept: Multiple variables; simple data relationships (comparisons, differences) · **Builds on:** Lesson 18 (a variable in a decision).

AI Connection (2 min): A model never relies on one number — it juggles accuracy, speed, examples per category, error. *First-principles question:* "Why can a single metric lie? Give an example where one number looks great but the system is bad."

Materials: LEGO CS & AI kit + device per pair, trigger objects/buttons, notebooks. **Model reuse:** Variables model from Lesson 16 — code only.

60-minute flow - Warm-up (8): "Two numbers, one judgment" — pairs decide a winner from two stats (points AND fouls) and defend it. Introduce reasoning over multiple variables. - **Concept + demo (10):** Demo two variables (score and a countdown timer), updating each, displaying both, and a condition that compares them (end when score > timer). - **Build/Code (30):** Pairs program at least two variables — a score counting up on events and a timer/count changing over a loop — then add a condition that compares or combines them (win/lose, lead by 3). They log both variables' meaning. *Swap at 15 min.* - **Challenge (7):** Compute and display a derived value (e.g., difference between the two) and react to it. - **Share + pack-up (5):** Pairs demo both variables changing and explain the relationship their program reasons about. Model stays built.

Pair roles: Each partner owns one variable; swap at 15 min to review the other's and co-write the comparison. (*Trio: third owns the derived value.*)

Differentiation: *Support* — one variable pre-built; students add the second. *Extension / fast-finisher* — three variables with a combined end condition. *Reteach* — build and test one variable fully, then copy the pattern for the second.

Exit check: Pair runs a program with two+ correctly labeled variables and a working condition that reasons about their relationship. **At-home (optional):** Pick a sport or game; list three stats it tracks and one judgment that needs two of them together. ~10 min.